

Semester	T.E. Semester V - Computer Engineering
Subject	Data Warehousing and Mining
Subject In-charge	Prof. Shweta Jaiswar
Laboratory	

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Experiment Number	01	
Experiment Title	Find the central tendency of the dataset by using statistics measures (mean median mode and range)	
CO Mapped	CO1	
Resources / Apparatus Required	Hardware: Computer system	Software: Python (Google Colab)
Description	- The terms mean, median and mode are used to describe the central tendency of a large data set. Range provides context for the mean, median and mode. - When working with a large data set, it can be useful to represent the entire data set with a single value that describes the "middle" or "average" value of the entire set. In statistics, that single value is called the central tendency and mean, median and mode are all ways to describe it. - To find the mean, add up the values in the data set and then divide by the number of values that you added. - To find the median, list the values of the data set in numerical order and identify which value appears in the middle of the list. - To find the mode, identify which value in the data set occurs most often. - Range, which is the difference between the largest and smallest value in the data set, describes how well the central tendency represents the data. If the range is large, the central tendency is not as representative of the data as it would be if the range was small.	

Program & Output	<pre> import pandas as pd import requests url = 'https://archive.ics.uci.edu/ml/machine-learning-databases/iris/iris.data' response = requests.get(url) with open('iris.csv', 'wb') as f: f.write(response.content) iris_df = pd.read_csv('iris.csv', header=None, names=['sepal_length', 'sepal_width', 'petal_length', 'petal_width', 'species']) def calculate_mean(data): return sum(data) / len(data) def calculate_median(data): sorted_data = sorted(data) n = len(sorted_data) if n % 2 == 0: mid1 = sorted_data[n // 2 - 1] mid2 = sorted_data[n // 2] return (mid1 + mid2) / 2 else: return sorted_data[n // 2] def calculate_mode(data): counts = {} for item in data: counts[item] = counts.get(item, 0) + 1 max_count = 0 mode = [] for item, count in counts.items(): if count > max_count: max_count = count mode = [item] elif count == max_count: mode.append(item) return mode def calculate_range(data): return max(data) - min(data) sepal_length_data = iris_df['sepal_length'].tolist() sepal_width_data = iris_df['sepal_width'].tolist() petal_length_data = iris_df['petal_length'].tolist() petal_width_data = iris_df['petal_width'].tolist() print("Sepal Length - Mean: (calculate_mean(sepal_length_data)), Median: (calculate_median(sepal_length_data)), Mode: (calculate_mode(sepal_length_data)), Range: (calculate_range(sepal_length_data))") print("Sepal Width - Mean: (calculate_mean(sepal_width_data)), Median: (calculate_median(sepal_width_data)), Mode: (calculate_mode(sepal_width_data)), Range: (calculate_range(sepal_width_data))") print("Petal Length - Mean: (calculate_mean(petal_length_data)), Median: (calculate_median(petal_length_data)), Mode: (calculate_mode(petal_length_data)), Range: (calculate_range(petal_length_data))") print("Petal Width - Mean: (calculate_mean(petal_width_data)), Median: (calculate_median(petal_width_data)), Mode: (calculate_mode(petal_width_data)), Range: (calculate_range(petal_width_data))") Sepal Length - Mean: 5.843333333333334, Median: 5.8, Mode: [5.0], Range: 3.6000000000000005 Sepal Width - Mean: 3.8400000000000003, Median: 3.0, Mode: [3.0], Range: 2.4000000000000004 Petal Length - Mean: 3.7586666666666666, Median: 4.35, Mode: [1.5], Range: 5.9 Petal Width - Mean: 1.1986666666666668, Median: 1.3, Mode: [0.2], Range: 2.4 import os if os.path.exists('iris.csv'): os.remove('iris.csv') </pre>
Conclusion :	The experiment successfully determined the Mean, Median, Mode, and Range on Mobile Price Prediction dataset using statistical formulas without using built-in functions. The experiment helped in understanding the concepts of central tendency and data dispersion. It was observed that mean, median, and mode represent the central value of the dataset, while range indicates the spread between the smallest and largest values. Thus, these statistical measures provide a simple way to summarize and interpret a dataset.